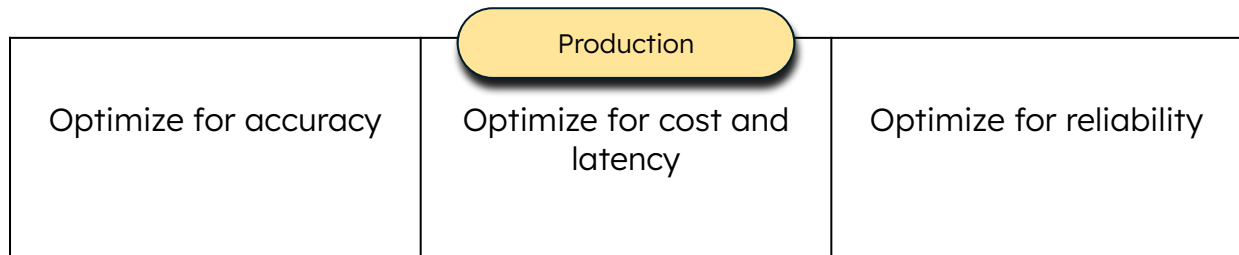
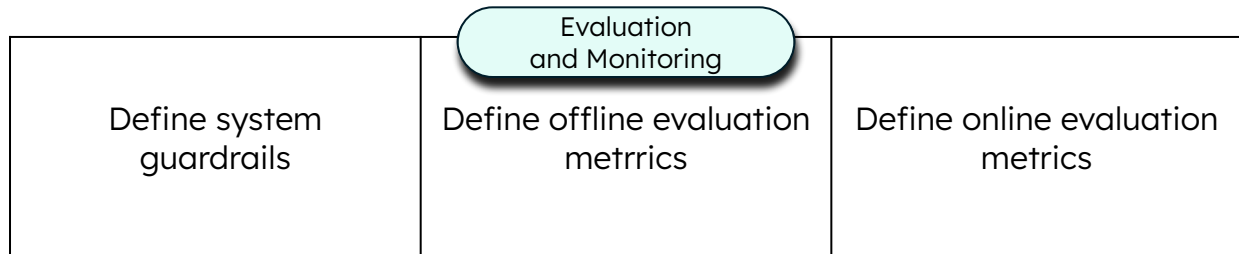
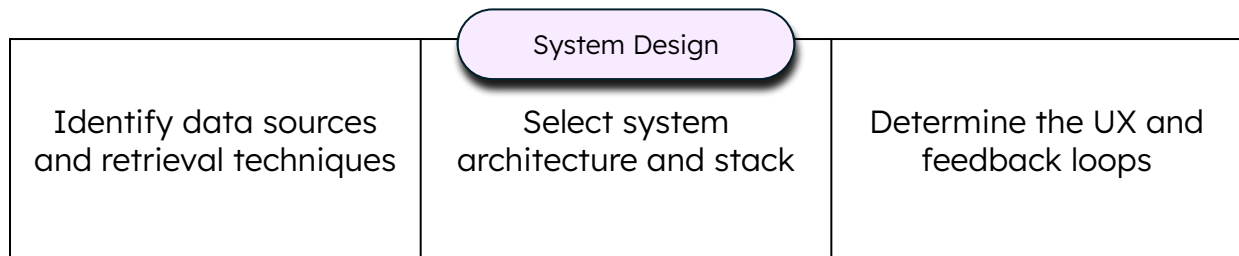
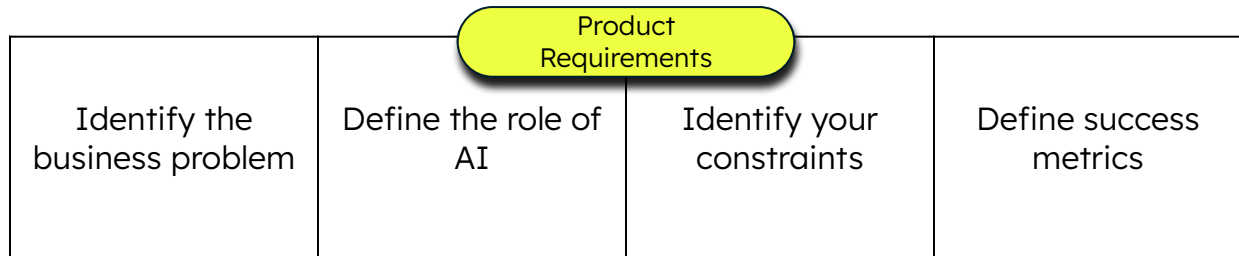


AI System Design Framework



Health Insurance Claims Review System

Background

Health insurers and health funds around the world employ medical reviewers to assess whether a requested treatment, procedure, or medication is covered under a patient's policy. This requires cross-referencing clinical documentation, coverage policies, clinical guidelines, and patient claims history. It is one of the most administratively intensive roles in healthcare operations.

What we are building

In this workshop, we are building an internal claims review system for a fictitious health insurance company called MDB Health. This system is used by medical reviewers to assess and adjudicate claims. The external-facing system through which healthcare providers submit claims and receive feedback on outcomes is out of scope.

Business Constraints

1. All denial decisions must include a human reviewer.
2. Reviewers must be able to override any system recommendation.
3. Complex cases, as defined by MDB Health's internal clinical guidelines, must be reviewed by a senior physician.
4. Patient data can only be processed within MDB Health's approved cloud environment.
5. Only models available on MDB Health's approved cloud provider are permitted.
6. All decisions must be auditable and traceable to a specific source document.

Performance Requirements

1. P95 response time for a recommendation must be under **10 seconds**.
2. System must handle up to **5,000 requests per day**.
3. Denial decisions must be communicated to the healthcare provider within **24 hours**.
4. Monthly model (embedding, LLM) costs must not exceed **\$10,000**.
5. **99.9%** uptime SLA required

Exercise 1. Product requirements

Business Problem

Medical reviewers at MDB Health spend an average of **2 days** processing claim review requests — **4 times** the industry standard for non-urgent cases and **12 times** the industry standard for urgent ones. Delays at this scale postpone patient care and are a leading cause of reviewer burnout across the organization.

Define the role of AI

Role	Answer
Critical / Complementary	
Reactive / Proactive	
Level of autonomy	

Define success metrics

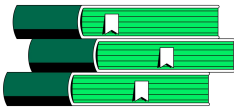
Instructions:

Define one SMART success metric for the MDBHealth claims review system that reflects the core business problem.

S: Specific, **M:** Measurable, **A:** Achievable, **R:** Relevant, **T:** Timebound



Exercise 2. Data and retrieval



Instructions:

The data sources that are available for you to use to build the system are listed below. Identify the update frequency, processing needs and retrieval techniques for each of the data sources.

Data source	Where does it live?	Format
Clinical guidelines	Confluence	PDF
Internal coverage policies	Confluence	PDF
Patient claims history	MongoDB	BSON

💡 Discuss what additional data sources might improve the system's recommendations.

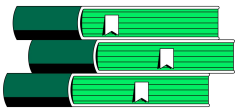
Estimate data update frequency

Instructions:

Estimate the update frequency of the data sources. Here are some options: Real-time, hourly, daily, monthly, quarterly, yearly, on-demand.

Data source	Update frequency
Clinical guidelines	
Internal coverage policies	
Patient claims history	

Exercise 2. Data and retrieval



Determine data processing needs

Instructions:

Determine the data processing needs of the data sources. Write down whether the data source needs ingestion-time or inference-time data processing, and a short note on the type of processing.

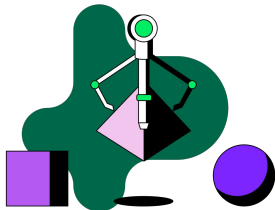
Data source	Data processing
Clinical guidelines	
Internal coverage policies	
Patient claims history	

Determine retrieval techniques

Instructions:

Determine the retrieval technique that best suits each data source. Choose one among: vector search, exact match, full-text search, hybrid search, or something else. If your response is “something else”, note down what that looks like.

Data source	Retrieval technique
Clinical guidelines	
Internal coverage policies	
Patient claims history	



Exercise 3. System architecture

Trace a claim from submission to decision

Instructions:

A claim has been received by MDB Health’s claims review system. In 5-7 steps, describe how the claim is reviewed and adjudicated. The outcome should be an approval or rejection.

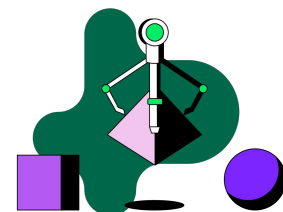
1. Claim request and clinical notes are received by the claims review system.

Identify the design patterns

Instructions:

Based on the the system flow, what design patterns would you use to implement this system?

Design pattern	Why?
RAG	Need to retrieve clinical guidelines, coverage policies and past claim history to inform the decision.



Exercise 3. System architecture

Define UX and feedback mechanisms

Instructions:

Think about the interface between the MDB Health claims review system and its users. Based on all the information you have so far, answer the following questions.

Remember, this system is for MDB Health medical reviewers, not the healthcare providers submitting claims or the patients whose care is being reviewed.

Decision	Answer
What does the system take as input?	
What does the system produce as output?	
Where does the system live?	
What triggers the system?	
What is the human's role?	
How does the system explain itself?	
How do users give feedback?	

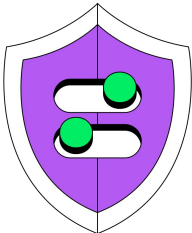
Exercise 4. Evaluation and monitoring

Design system guardrails

Instructions:

Define what are acceptable inputs and outputs to the system. Come up with 2 input and output guardrails.

Type	Guardrail
Input	Claims should have supporting clinical documentation
Input	
Input	
Output	Final result should provide citations
Output	
Output	



Exercise 4. Evaluation and monitoring

Define offline evaluation metrics

Instructions:

Define the system metrics you would use to evaluate the system before deploying it to production.

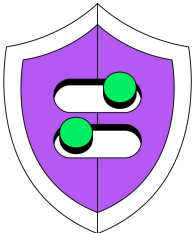
Type	Metric
Guardrail compliance (input/output)	Claim rejection rate, Hallucination rate
Response quality	
Domain-specific accuracy	
Performance	

Define online evaluation metrics

Instructions:

Define the additional metrics you would use to monitor the system in production.

Type	Metric
User feedback	
Behavioral	



SOLUTION

Exercise 1. Product requirements

Business Problem

Medical reviewers at MDB Health spend an average of **2 days** processing claim review requests — **4 times** the industry standard for non-urgent cases and **12 times** the industry standard for urgent ones. Delays at this scale postpone patient care and are a leading cause of reviewer burnout across the organization.

Define the role of AI

Role	Answer
Critical / Complementary	Complementary
Reactive / Proactive	Reactive
Level of autonomy	Semi-autonomous

Define success metrics

Instructions:

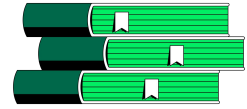
Define one SMART success metric for the MDBHealth claims review system that reflects the core business problem.

S: Specific, **M:** Measurable, **A:** Achievable, **R:** Relevant, **T:** Timebound

Reduce the average processing time for urgent claim review requests from 2 days to 4 hours within 90 days of launch.



Exercise 2. Data and retrieval

**Instructions:**

The data sources that are available for you to use to build the system are listed below. Identify the update frequency, processing needs and retrieval techniques for each of the data sources.

Data source	Where does it live?	Format
Clinical guidelines	Confluence	PDF
Internal coverage policies	Confluence	PDF
Patient claims history	MongoDB	BSON



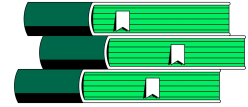
Discuss what additional data sources might improve the system's recommendations.

Estimate data update frequency

Instructions:

Estimate the update frequency of the data sources. Here are some options: Real-time, hourly, daily, monthly, quarterly, yearly, on-demand.

Data source	Update frequency
Clinical guidelines	On-demand
Internal coverage policies	On-demand
Patient claims history	Real-time



Exercise 2. Data and retrieval

Determine data processing needs

Instructions:

Determine the data processing needs of the data sources. Write down whether the data source needs ingestion-time or inference-time data processing, and a short note on the type of processing.

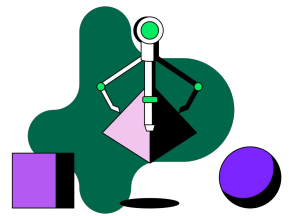
Data source	Data processing
Clinical guidelines	Ingestion-time
Internal coverage policies	Ingestion-time
Patient claims history	None

Determine retrieval techniques

Instructions:

Determine the retrieval technique that best suits each data source. Choose one among: vector search, exact match, full-text search, hybrid search, or something else. If your response is “something else”, note down what that looks like.

Data source	Retrieval technique
Clinical guidelines	Hybrid search
Internal coverage policies	Hybrid search
Patient claims history	Exact match



Exercise 3. System architecture

Trace a claim from submission to decision

Instructions:

A claim has been received by MDB Health's claims review system. In 5-7 steps, describe how the claim is reviewed and adjudicated. The outcome should be an approval or rejection.

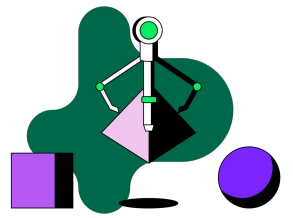
1. Claim request and clinical notes are received by the claims review system.
2. Patient's claims history, and relevant clinical guidelines and coverage policies are retrieved.
3. An LLM generates a coverage recommendation.
4. If it is a complex case, escalate the recommendation to a senior physician.
5. If the recommendation is a denial, escalate it to a medical reviewer.
6. Log the final decision along with the reasoning

Identify the design patterns

Instructions:

Based on the the system flow, what design patterns would you use to implement this system?

Design pattern	Why?
RAG	Need to retrieve clinical guidelines, coverage policies and past claim history to inform the decision.
LLM-as-a-router	To classify incoming claims by complexity and route to the appropriate workflow.
Human-in-the-loop	Denial decisions and complex cases need to be reviewed by a human.



Exercise 3. System architecture

Define UX and feedback mechanisms

Instructions:

Think about the interface between the MDB Health claims review system and its users. Based on all the information you have so far, answer the following questions.

Remember, this system is for MDB Health medical reviewers, not the healthcare providers submitting claims or the patients whose care is being reviewed.

Decision	Answer
What does the system take as input?	Claim request form, clinical notes
What does the system produce as output?	Approval or denial verdict along with an explanation
Where does the system live?	Embedded within MDB Health's existing claims management system
What triggers the system?	A claim submission
What is the human's role?	Review the system's assessment and make the final decision
How does the system explain itself?	By citing the clinical guidelines and coverage policies that informed the assessment
How do users give feedback?	By overriding the verdict and logging a reason, flagging irrelevant citations

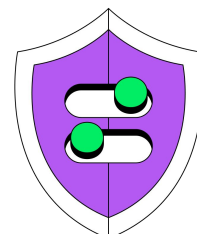
Exercise 4. Evaluation and monitoring

Define system guardrails

Instructions:

Define what are acceptable inputs and outputs to the system. Come up with 2 input and output guardrails.

Type	Guardrail
Input	Claims should have supporting clinical documentation
Input	
Input	
Output	Final result should provide citations
Output	Autonomous denials are not allowed
Output	Final response should not contain incorrect clinical guidelines or policy references



Exercise 4. Evaluation and monitoring

Define offline evaluation metrics

Instructions:

Define the system metrics you would use to evaluate the system before deploying it to production.

Type	Metric
Guardrail compliance (input/output)	Claim rejection rate, hallucination rate
Response quality	Citation relevance, faithfulness
Domain-specific accuracy	Override rate
Performance	Average processing time

Define online evaluation metrics

Instructions:

Define the additional metrics you would use to monitor the system in production.

Type	Metric
User feedback	Time to decision
Behavioral	Average review time, auto-approval rate

